Document Title

Your Name (Student ID)

July 21, 2026

1 Text Formatting

This sentence has **bold text**, *italic text*, *emphasized text*, and underlined text. SMALL CAPS and `typewriter` also show up occasionally.

Font sizes: small, normal, large, Large, LARGE, huge, Huge.
Colored text in red and colored text in blue are common for highlighting contrasted statements.

2 Lists

- Primary item
- Secondary item
    1. Nested numbered item
    (a) Custom-labeled sub-item

    - Dash-labeled item
        † Symbol-labeled item (dagger)
        ⋄ Symbol-labeled item (diamond)
        ▷ Symbol-labeled item (triangle)

(I) Roman-numbered item

(II) Another roman-numbered item

Bold-run-in label description text continues on the same line;
a manual line break is used above to keep the next line indented under it.

3 Math

Inline math uses `$...$`; display math uses `$$...$$`, `\[...\]`, or the numbered `equation` environment.

Superscripts/subscripts: x^2, x_i, x_{i_2} (always brace nested subscripts). Greek: $\alpha, \beta, \gamma, \delta, \pi, \Pi, \sigma, \theta, \omega$. Fractions: $\frac{a}{b}$ vs. $\dfrac{a}{b}$ (display-style, taller); inline $\dfrac{a}{b}$ forces the same taller look mid-paragraph. Roots: $\sqrt{2x+1}$, $\sqrt[5]{2x+1}$. Sized brackets: $\left(\dfrac{1}{x^2+1}\right)$, $[\cdot]$, $\{\cdot\}$, $|\cdot|$, one-sided: $\left.\dfrac{dy}{dx}\right|_{x=1}$. Sums/limits/integrals: $\sum_{i=1}^{n} x_i$, $\lim\limits_{x \to a} f(x)$, $\int_0^\infty e^{-st} f(t)\, dt$. Sets/logic: $\mathbb{R}, \mathbb{Z}, \mathbb{Q}, \mathbb{N}$, $\forall a \in \mathbb{R}$, $\exists$.

$$L = I_{ij} \cdot \cos(\theta) + R^2 \tag{1}$$

$$\begin{split} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{split} \tag{2}$$

$$\begin{align} x^2 + 4 &= 4x \tag{3} \\ (x-2)^2 &= 0 \tag{4} \\ x &= 2 \tag{5} \end{align}$$

$$p(t) = \begin{cases} 0, & t < 0, \\ e^{-t}, & 0 \leq t < 3, \\ \ln(t-2), & t \geq 3 \end{cases}$$

Equation (1) and Equation 1 are the `\eqref` and `\ref` forms respectively – `\eqref` auto-adds the parentheses.

4 Tables

Method	Accuracy	Time Complexity
Rasterization	Medium	$O(n)$
Ray Tracing	High	$O(n^2)$

Table 1: Plain bordered table

<table>
<tr><th rowspan="2">Family</th><th colspan="2">Equation</th><th rowspan="2">Condition</th></tr>
<tr><td>Time function</td><td>Transform</td></tr>
<tr><td rowspan="2">Algebraic</td><td>1</td><td>$\frac{1}{s}$</td><td>$s > 0$</td></tr>
<tr><td>t</td><td>$\frac{1}{s^2}$</td><td>$s > 0$</td></tr>
</table>

Table 2: Multirow + multicolumn table with partial cline rules

Term	**Description**
$f(x)$	A long description that needs to wrap onto multiple lines, hence the `p{5cm}` column type instead of `c`.

Table 3: Colored table using xcolor[table] (`rowcolor`/`cellcolor`)

5 Figures

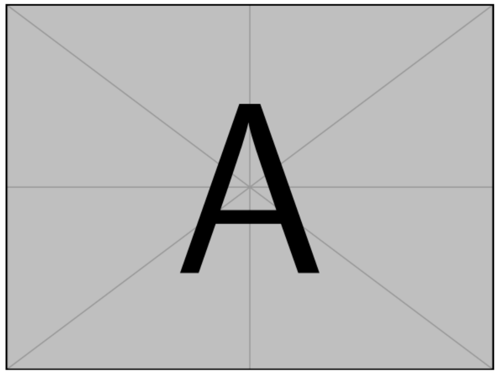
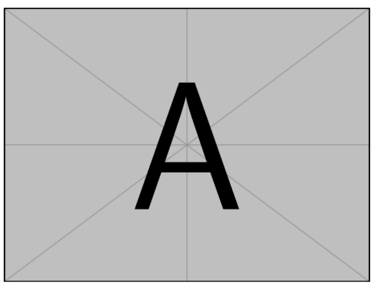
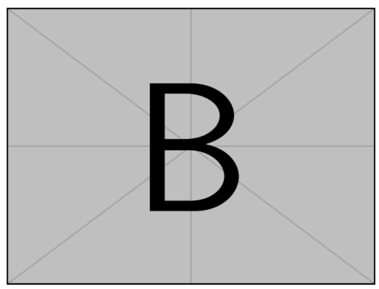


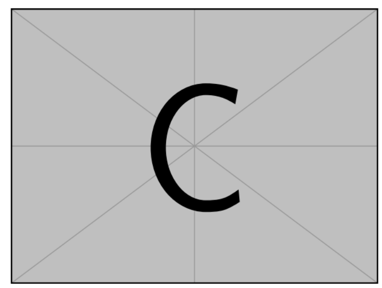
Figure 1: A single figure



(a) Output A

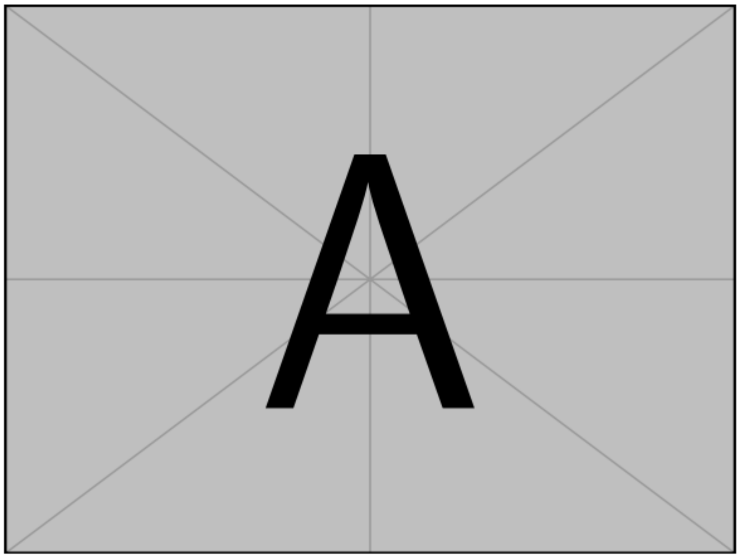


(b) Output B

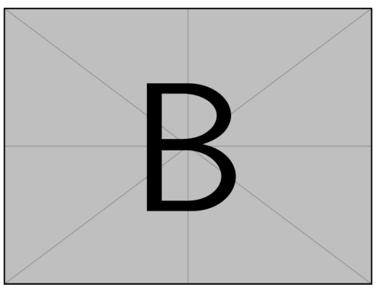


(c) Output C

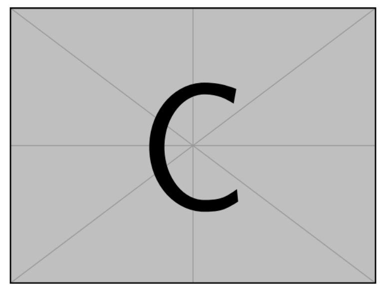
Figure 2: Three subfigures in a single row, each with its own sub-caption



(a) Main diagram



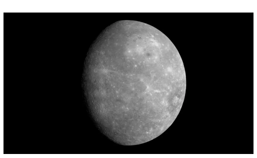
(b) Detail B



(c) Detail C

Figure 3: Asymmetric layout: one dominant figure above two smaller ones

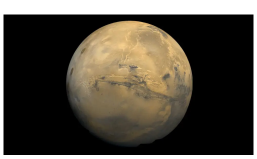
Row Label One



(a) Mercury

(b) Venus

Row Label Two



(c) Mars

(d) Jupiter

Figure 4: Grid of subfigures split into labeled row-groups

Text placed after a `wrapfigure` environment flows around the image instead of stacking above/below it, which is the key visual difference from a plain `figure`. Use `{r}` or `{l}` for the side, and a width fraction for how much horizontal space the wrap reserves. Keep enough body text below the environment to fill the wrapped height, otherwise later content can overlap the image.

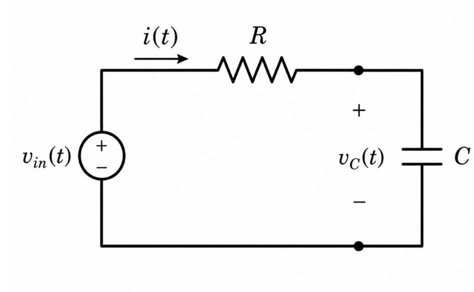


Figure 5: Wrapped figure

6 Code Listings

Listing 1: A small Python function.

```
def greet_user(name):
    # Create a greeting message using an f-string
    message = f"Hello, {name}! Welcome to Python."

    # Print the greeting message
    print(message)

    # Return the message in case it's needed elsewhere
    return message


# Example usage
user_name = "Alice"
greet_user(user_name)
```

7 Footnotes, Hyperlinks & Citations

A footnote can be attached to any word.[1] A raw link: `https://www.overleaf.com`. A labeled link: Overleaf.

Citations use natbib: [1]. This resolves to a real number (not `[?]`) because of the `thebibliography` block near the end of this file – run pdflatex *twice* (or use latexmk) so the `.aux` data from the first pass is available to resolve it on the second.

In your actual solves, the bibliography instead comes from an external `.bib` file via `\bibliographystyle{...}` + `\bibliography{filename}` (no `.bib` extension), which needs a bibtex/biber run in addition to pdflatex, e.g. a file named `refs.bib` containing:

```
@book{sample2024,
    author    = {Author Name},
    title     = {Book Title},
    publisher = {Publisher},
    year      = {2024}
}
```

Common styles: `plain`, `ieeetr`, `unsrtnat`, `apalike`.

8 Theorem and Proof

Proposition 1 (Sample proposition)**.** *For a circular orbit of radius* r, $\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2}$.

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$; divide by c^2. □

See Proposition 1 for the cross-reference form of a theorem.

References

[1] Author Name. *Book Title*. Publisher, 2024.

[1] This is a footnote.